

Brian M. Howell

Distributed Systems

| Infrastructure Architecture

| Numerical Computing

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∈ Bay Area, CA

Summary

Computational systems architect with a record of designing enterprise-scale platforms, standardizing distributed computing, and aligning cross-org strategy. Drives HPC/ML platform adoption across orgs, reducing cost and accelerating workflows.

Experience

Apple, Cupertino, CA

Feb 2024 – present

Computational Engineer, Platform Architecture

- **Enterprise Orchestration Platform:** Architected and rolled out an orchestration system adopted by 5+ engineering organizations. Standardized distributed computing practices, enabling 1m+ job executions with automated failure recovery and adaptive scheduling, reducing execution time from weeks to minutes.
- **Scalable Computing Infrastructure:** Enhanced computational solvers to efficiently scale across heterogeneous compute architectures including multi-GPU clusters and distributed environments. Developed optimizations enabling previously infeasible large-scale workloads while maintaining numerical accuracy and stability.
- **AI Infrastructure & Training Systems:** *Currently* building scalable data generation and distributed training pipelines for ML workflows. Automated dataset creation and training cycles, enabling rapid experimentation and faster model iteration.
- **Mentorship & Talent Development:** Mentored 3 PhD interns on HPC-based projects, guiding integration of research prototypes into production infrastructure and accelerating project delivery.

Google X, Mountain View, CA

Jan 2022 – Dec 2022

AI Resident, Early-Stage Project

- **Computational Optimization & Strategy:** Directed development of optimization frameworks integrating machine learning to accelerate experimental workflows. Implemented Bayesian optimization and statistical models to guide autonomous decision-making in complex parameter spaces.
- **Forward Simulation Integration:** Led integration of predictive models with forward simulations to guide autonomous experimental design, reducing iteration cycles and enabling data-driven strategic decisions.
- **Technical Innovation:** Contributed to three patent applications in computational optimization and autonomous systems, translating research innovations into protected intellectual property with potential organizational impact.

Lawrence Livermore National Lab., Livermore, CA

Jun 2017 – Jan 2022

Staff Scientist, Material Dynamics & Kinetics

- **Systems Architecture & Digital Infrastructure:** Shaped digital twin frameworks, forward simulation models, and closed-loop optimization systems for precision manufacturing, integrating software and hardware pipelines to enable cross-functional R&D efficiency.
- **Autonomous Control Systems:** Designed development of intelligent controllers, in-situ monitoring systems, and process optimization algorithms for real-time operations, establishing foundations for adaptive manufacturing strategies.
- **Instrumentation & Process Development:** Led integration of sensor systems and design of experimental protocols for manufacturing characterization, standardizing analytical workflows to improve R&D throughput.
- **Research Impact:** Delivered two patent applications and peer-reviewed publications, positioning research outputs to influence broader organizational technology strategy.

Skills

Systems & Architecture: Computational Optimization, Forward Simulation, Distributed Infrastructure, Parallel Computing, Systems Integration, Enterprise Platforms

Programming & Tools: C/C++, Python, GPU Computing, HPC Systems

Technical Strategy & Leadership: Algorithm Design, Cross-functional Technical Leadership, Technical Program Management

Education

UC Berkeley

PhD/MS in Computational/Mechanical Engineering

2019–2024

Dissertation: *Simulation-Informed Optimization and Machine Learning for Advanced Manufacturing*

Minors: *Math, Optimization*

Brigham Young University

BS in Chemical Engineering

2013–2017